Assignment: Network Security Topology Design Using a Variety of Products

Objective: In this assignment, you will design and implement a secure network topology using a variety of network security services using various technologies of network devices. The goal is to provide a real-world, multi-vendor environment where you can apply network security best practices.

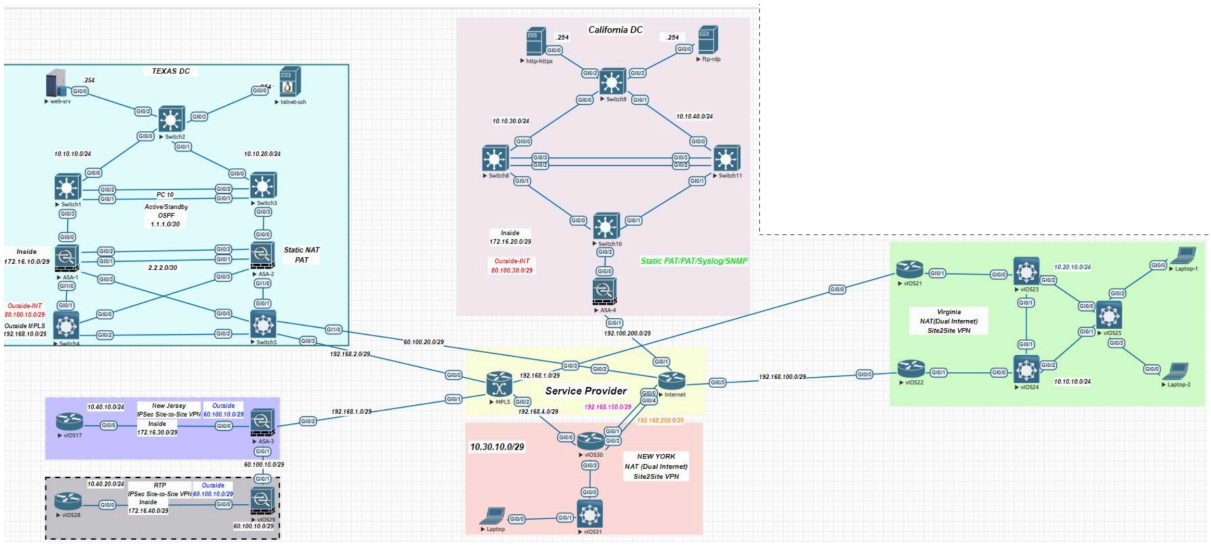
This assignment will help you gain practical experience in designing and implementing a secure network using a mix of industry-standard technologies and products. By completing this task, you'll build a solid foundation in network security concepts that are applicable in real-world scenarios.

Scenario Overview: Imagine that you are tasked with designing the network for a small-to-medium-sized business (SMB) that operates in a hybrid cloud environment. The business has multiple departments, including HR, Finance, Sales, and IT. Each department has its own set of sensitive data, and the company requires secure communication both internally (within the local area network) and externally (to/from the internet and the cloud).

Your task (in a group of 4 person per group) involves configuring various security policies, such as Access Control Lists (ACL), Network Address Translation (NAT), Virtual Private Network (VPN) and potentially an Intrusion Prevention System (IPS).

Optimizing firewall policies is also crucial, defining rules and actions to govern traffic passing through your firewall devices. These policies dictate which traffic is permitted, denied, or logged, ensuring a robust security posture for your network infrastructure. Understanding the intricacies of these components and their interactions is essential for effectively securing and managing your network environment. (Ah M., April 2024)

Requirements:



1. **Topology Design**:

You will need to create a secure network topology that ensures data confidentiality, integrity, and availability. In this topology guideline (you may follow this topology or create your own), you need to have these network devices for crucial perimeter defence:

- **ASAs (Adaptive Security Appliance)** serving as primary firewalls and security devices as well as VPN connectivity.
- **MPLS Router** for efficient packet forwarding and routing which ensure smooth data transmission across network infrastructure.
- **Internet Router** performing essentials functions such as NAT, routing and secure external networks.

2. **Network Segmentation**:
    - Create VLANs to isolate different departments (HR, Finance, IT, and Sales).
    - Use private IP addressing for internal communication, and public IP addresses as needed for internet-facing devices.
    - Implement access control lists (ACLs) to limit traffic between VLANs based on the business needs (e.g., Finance cannot communicate with IT, but HR can communicate with Sales).
3. **Firewall Configuration**:
    - Configure a firewall to protect the perimeter of the network. Ensure that only authorized services are allowed in and out of the network.

- Use different types of firewall rules (stateful, application-layer filtering, etc.) for network traffic security.
- Use security zones (e.g., internal, DMZ, external) for proper segmentation and traffic control.

4. **VPN for Remote Access**:
   - Implement VPN technology to allow secure remote access for employees (SSL VPN or IPsec VPN).
   - If using cloud services (e.g., AWS or Azure), configure a VPN tunnel between the on-premises network and the cloud to ensure secure communication between the data center and cloud resources.
5. **IDS/IPS Implementation**:
   - Place an IDS/IPS between the internal network and the internet, or between your network and any sensitive areas (e.g., finance or HR data).
   - Configure the IDS/IPS to detect and prevent common network attacks like DoS, SQL injection, or unauthorized access attempts.
6. **Secure Communication**:
   - Configure encrypted communication (e.g., IPSec, SSL/TLS) between devices, ensuring that sensitive data is protected in transit.
   - If appropriate, use secure email or web services for communication (e.g., HTTPS for web traffic, SMTPS for email).
7. **Documenting the Network Design**:
   - Provide a **detailed network diagram** showing the placement of devices (routers, switches, firewalls, IDS/IPS, etc.).
   - Label the subnets, VLANs, IP addresses, and indicate the type of security measures implemented on each device.
   - Provide a brief description of the role and purpose of each device in the network.
8. **Security Policies and Procedures**:
   - Define security policies (e.g., password policies, device hardening procedures, etc.) that need to be applied to each device.
   - Discuss how each policy helps in maintaining the security of the network.
9. **Evaluation and Testing**:
   - After configuring the devices and applying the security measures, simulate traffic in the network (e.g., using a packet sniffer or network simulation tool like GNS3, Cisco Packet Tracer, or Wireshark).

- Verify that the security configurations are working as expected. For example, test if VPN connections are secure, if the firewall blocks unauthorized traffic, and if the IDS/IPS is detecting attacks.

Deliverables:

1. **Network Topology Diagram**: Include a clear and labelled diagram of the entire network topology as well as network segmentation (IP Addresses and Subnet Mask)
2. **Configuration Files**: Provide the configuration files for routers, switches, firewalls, IDS/IPS, and other devices used.
3. **Security Policy Documentation**: Submit a document detailing the security policies applied to each device (password policies, access control lists, firewall rules, etc.).
4. **Testing and Evaluation Report**: Include a report on how you tested the security measures and the results.
5. **Executive Summary**: Write a 1-2 page summary explaining your network design choices, security configurations, and how each measure helps protect the network.

Tools and Resources:

- **Cisco Packet Tracer** or **GNS3** for simulating Cisco-based networks.
- **Eve-ng Web Login** (another option for simulation tools)
- **Wireshark** for packet analysis.
- **VirtualBox** or **VMware** for running virtualized devices (e.g., firewalls, IDS/IPS).
- Documentation on the devices/products you are using (Cisco, Fortinet, Palo Alto, etc.)

Grading Criteria:

- **Network Topology and Design**: Is the network segmented properly? Are the devices chosen appropriate for the scenario? **[10 marks]**
- **Security Measures**: Are the security measures (firewalls, IDS/IPS, VPNs, etc.) configured correctly? Are the policies effective? **[15 marks]**
- **Documentation and Clarity**: Is the network diagram clear and well-labeled? Are the configurations and policies explained in detail? **[10 marks]**
- **Testing and Evaluation**: Does the testing show that security features work as intended? **[15 marks]**

- **Teamwork**: Does everybody do their parts? Is everybody showing up during the assignment consultation? **[10 marks]**

TOTAL MARKS: 60 MARKS